

Tommaso Alfonsi

Senior AI & Full-Stack Engineer

Lombardy, Italy & Canton of Ticino, Switzerland	tommaso.alfonso@gmail.com	https://tomalf2.github.io
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Summary	Software Engineer and Researcher with 7 years of combined experience in designing, developing, and deploying complex software solutions, full-stack architectures, and applied artificial intelligence. Specialized in engineering RAG/LLM pipelines, software security controls (SAST, DAST, SCA), and web applications for the industrial and aerospace sectors. Solid background in computational genomics from a PhD and PostDoc at Politecnico di Milano, with a proven ability to translate complex scientific and algorithmic challenges into production-ready, scalable software.
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Core Competences	• AI & Machine Learning: RAG Pipelines, Large Language Models (LLM), Vector Databases (Qdrant), Fine-tuning, Generative AI • Full-Stack & Web Engineering: Python (FastAPI, Flask, Streamlit), JavaScript/TypeScript (Vue.js), REST APIs, Microservices • Cybersecurity & DevSecOps: Software Security Controls (SCA, SAST, DAST), CI/CD Pipelines, Docker, Containerization, Cloud (AWS/Azure) • Data Engineering & Orchestration: ETL Pipelines, Dagster, PostgreSQL, MongoDB, High-Performance Computing (HPC). Links and insights into my industrial and academic projects: https://tomalf2.github.io/projects/
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Professional and R&D Experience	Full-Stack & AI Software Engineer <i>MotusML s.r.l. 2025 – present</i> • Designed and implemented end-to-end RAG (Retrieval-Augmented Generation) pipelines using Vector Databases (Qdrant), converting aerospace documentation and technical reports into high-value structured assets for internal AI systems.. • Orchestrated and automated complex data workflows and pipelines using Dagster, ensuring high reliability, monitoring, and software resilience. • Developed full-stack industrial web applications (Vue.js, FastAPI, Flask) for real-time monitoring and control of production processes. • Integrated and automated code cybersecurity controls (SCA, SAST, DAST) into corporate CI/CD pipelines, elevating software security and compliance standards. • Managed the full software development lifecycle: from requirement analysis and task definition to containerized deployment on corporate networks. • Prepared project deliverables and presented progress reports (Milestones/SAL) to external clients. • Mentored junior developers and conducted pair-programming sessions to enhance internal team capabilities.
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Senior R&D Software Engineer & Data Scientist (PhD & PostDoc)

Politecnico di Milano – Computational Genomics / 2020 – 2025

- Designed and developed **custom libraries, APIs, and interactive dashboards** for analyzing and visualizing complex, high-dimensional genomic data.
- Engineered high-performance **ETL pipelines** and algorithms, deployed across cloud/HPC **distributed computing** clusters.
- Technically **coordinated international research projects, managed data lifecycle**, attended international **conferences**, and **published findings** in high-impact scientific journals (**Nature Communications, Science Advances**, Nucleic Acids Research, BMC Bioinformatics, etc.).
- **Academic Teaching & Mentorship**: Conducted practical laboratory lectures for Master's degree courses and mentored graduate students in coding, software development, and data modeling.

Languages

English
Fluent – (TOEIC 870/990 ~C1)

Italian
Native

Education

2017 - 2019

MSc, Computer Science Engineering
Politecnico di Milano, Milan, Italia

2020 - 2023

PhD, Information Technology,
Politecnico di Milano, Milan, Italia

Awards

- Recipient of the Dimitri Chorafas Award for outstanding scientific achievements and doctoral dissertation.

Soft Skills & Leadership

- **Technical Communication & Relationship Management**: Proven ability to translate complex architectural and scientific concepts for both technical and non-technical stakeholders, liaising effectively with clients, project managers, and engineering teams.
- **Leadership & Mentorship**: Collaborative mindset, strong focus on code reviews, pair-programming, and fostering team members' growth.
- **Ownership & Delivery**: Results-oriented end-to-end mindset, balancing release velocity, code security, performance, and architectural scalability.

Personal Details & Contacts

Based in: Varese, Lombardy, Italy
Nationality: Italian

Date of Birth: December 29, 1993

Open to full-remote, hybrid, or on-site opportunities in Northern Italy and Switzerland.

Email: tommaso.alfio@gmail.com

Portfolio: <https://tomalf2.github.io>



I look forward to discussing how my skills and experience can contribute to your team.